

Experiment ‘p_6_1_c_process_array_reverse_order_seq’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_1_c_process_array_reverse_order_seq

Natural language statement: Write a method that takes an array of ten integers and then returns all elements in reverse order.

Method signature: p_6_1_c_process_array_reverse_order_seq(inputs: seq<int>) returns (reversed: seq<int>)

Ensures

- `|reversed| == |inputs|`
- `forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i]`

Requires

- `|inputs| == 10`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes an array of ten integers and then returns all elements in
→ reverse order.

The signature should be:

p_6_1_c_process_array_reverse_order_seq(inputs: seq<int>) returns (reversed: seq<int>)

The method should respect the following contract:

requires |inputs| == 10, ensures |reversed| == |inputs|, ensures forall i :: 0 <= i < |
→ reversed| ==> reversed[i] == inputs[|inputs| - 1 - i]

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn08KII2ax6K8PL4T0zlkY8DgcCcd

Model: gpt-5.5-2026-04-23

Created at: 1780570956

Finish reason: stop

Usage: 358 tokens in, and 564 tokens out

```
{  
  reversed := seq(|inputs|, i requires 0 <= i < |inputs| => inputs[|inputs| - 1 - i]);  
}
```

Final Program

```
method p_6_1_c_process_array_reverse_order_seq(inputs: seq<int>) returns (reversed: seq<  
→ int>)  
  requires |inputs| == 10  
  ensures |reversed| == |inputs|  
  ensures forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i  
→ ]  
{  
  reversed := seq(|inputs|, i requires 0 <= i < |inputs| => inputs[|inputs| - 1 - i]);  
}
```

Total Token Usage

Input tokens: 358

Output tokens: 564

Reasoning tokens: 512

Sum of 'total tokens': 922

Experiment Timings

Overall Experiment started at 1780570955642, ended at 1780570993800, lasting 38158ms (38.16 seconds)

Iteration #1 started at 1780570955647, ended at 1780570993799, lasting 38152ms (38.15 seconds)